Comparative Evaluation of Indirect Pulp Therapy in Young Permanent Teeth using Biodentine and Theracal: A Randomized Clinical Trial

Bushra Rahman*/ Mousumi Goswami**

INTRODUCTION

Dental caries is one of the most frequently occurring diseases affecting human beings that becomes a challenge for dentists as the disease progresses. The ultimate goal of operative and endodontic treatment is to preserve and maintain the vitality of pulp so as to allow continued development of odontogenic apparatus.[1]

The discussion regarding the extent of caries excavation in order to arrest the caries process in a deep carious lesion is not new.[2] John Tomes (1859) wrote "it is better that a layer of discolored dentine be allowed to remain for the protection of the pulp rather than run the risk of sacrificing the tooth".[2]

In a primary / permanent tooth with deep dentinal caries; judiciously removing the infected dentin and isolating the affected dentin from oral fluids with suitable material is referred to as indirect pulp therapy (IPT). IPT has become the front runner in vital pulp therapy in recent years and is a procedure performed in teeth with deep carious lesions approximating the pulp but without any signs or symptoms of irreversible pulpal changes. Peripheral caries along with soft caries on pulpal floor is removed while the caries on cavity floor which is closest to the pulp is not excavated so as to avoid pulp exposure. It is then covered with a suitable biocompatible material.[3]

From the Department of Pediatric and Preventive Dentistry,I.T.S Dental College and Hospital.
*Bushra Rahman, Lecturer.
** Mousumi Goswami Professor and Head of Department

Corresponding author:
Mousumi Goswami
Department of Pediatric and Preventive Dentistry
I.T.S Dental College and Hospital, 47 Knowledge Park –III
Greater Noida, U.P, India
Phone:+ 8588854190
E-mail: mousumi_leo@yahoo.co.in

In the search for an "ideal" material, a wide range of restorative materials have been introduced over the past years and many have been discontinued. The most popular material is Calcium hydroxide which is also considered as the gold standard for IPT because of its beneficial properties such as high pH, low cytotoxicity and induction of mineralization. However, the poor dentinal bonding, mechanical instability, dissolution over time leading to formation of tunnels in the dentinal barrier are few disadvantages.[4] In this context, an interesting development was the introduction of calcium silicate based cements .[5] Biodentine® also known as "dentin in capsule", is one new formulation which is a biocompatible and bioactive dentin substitute that was developed as a pulp capping agent.[5] It has attracted the attention due to its fast setting time, high compatibility, high compressive strength, high pH (pH=12), excellent setting ability, ease of handling, rich source of calcium ions and remineralizing property as well as its versatile range of clinical applications in restorative dentistry, endodontics and dental trauma etc.[6,7]

One of the recently developed light curable resin –modified calcium silicate material is Theracal®. Its ability to get cured at a depth of 1.7mm decreases the chance of early dissolution with an improved sealing and bonding ability to moist dentin. These properties offer major advantage in pulp capping procedures.[8]

Thus, this study was aimed at evaluation of Biodentine, Theracal and Dycal as indirect pulp therapy agents in young permanent teeth.

MATERIALS AND METHOD

This randomized clinical trial was conducted in the Department of Pediatric and Preventive Dentistry between December 2016 and May 2019 using a protocol that was reviewed and approved by the Institutional Review Board and Ethical committee letter number (ITSCDSR/L/2018/136) and registered under CTRI (CTRI/2018/02/014782).

Sixty children aged 7-15 years who had at least one deeply carious permanent molar were selected for this study. The lesion was evaluated clinically and radiographically and diagnosed as deep dentinal caries approaching pulp without potential pulp exposure.

The parents / guardians of the patients were explained about the study and an informed consent was obtained.

Inclusion criteria

Permanent posterior teeth with deep dentinal caries (ICDAS 5 & 6) and open apices were included from patients who presented with clinical symptoms of sensitivity to cold and pain on mastication. Radiographically, the caries involvement of >70% or 2/3rd of dentin thickness without any presence of periapical pathology, pathologic external / internal root resorption or pulp calcification were included.

Sample size calculation

Considering type I error to be 5% (α – 0.05) and power equal to 80 % with 10% outcome difference and 10% dropout rate led to the required sample size of 45 which was divided into 3 equal groups of 15 each. The sample was increased to 20 in each group to improve the validity of the study as well as to compensate for possible follow up loss.

Sampling technique and Randomization

The study was carried out with the help of two investigators. The first investigator randomized the subjects in accordance with the CONSORT 2010 guidelines. Three different colored balls were coded for the three different intervening materials. Simple randomization was done by asking the patients to choose one of the three differently colored balls which would determine the intervening material to be placed for IPT. In case the patient had more than one tooth fulfilling the inclusion criteria, randomization was done separately for each tooth. The samples were divided into following three groups:

Group 1: Biodentine (Septodont, France)

Group 2: Theracal (BiscoInc, Schamburg, IL)

Group 3: Dycal (Dentsply Caulk Milford, DE, USA)

Once the sample size of one particular group was achieved, the colored ball representing that group was removed from further randomization process. The present study is a double blinded study. The participant was blinded from the intervening material used, the clinician doing the excavation was not blinded since different materials require different manipulating instructions and the investigator assessing the signs and symptoms was unaware of the intervening material placed.

Clinical procedures

All the procedures were carried out by one investigator who was trained and had conducted pilot study on a sample of 8 similar patients. Cold pulp testing was performed by placing a small cotton pellet sprayed with ENDO FROST (propane-butane mixture, Roeko, Germany) frozen gas (-50°C) on the buccal surface of tooth. The tooth was anesthetized and isolated using rubber dam. Removal of enamel caries was done using diamond fissured bur No. 330 mounted in high speed hand piece with copious water. Based on hardness, tactile sensation and visual criteria, the soft infected dentin layer was carefully removed with a sharp spoon excavator. Excavation was stopped when the dentin showed increased resistance, thus, leaving behind the hard affected dentin over the pulpal floor so that the pulp tissue was not exposed. A solution of 1% NaOCl in a syringe with needle was used to irrigate the cavity every 3 minutes in order to wash away dentinal debris. In case of an accidental pulp exposure, the tooth was excluded from the study.

After excavation, the remaining dentin layer was covered with corresponding randomly assigned intervening liner material manipulated according to manufacturer's instructions. This was followed by restoration with a layer of GIC and bulk restoration with resin composite.

Two investigators performed the clinical and radiographic evaluations at 3weeks, 3months, 6 months, 12 months, 18 months and 24 months.

Result analysis criteria

Result of IPT was analysed as success or failure according to clinical and radiographic criteria.

Clinical/ primary criteria included a record of the presence of postoperative pain, sensitivity to palpation and percussion, mobility and a visual examination of surrounding soft tissues for the presence of any inflammation, sinus tract or fistula.

Radiographic/ secondary criteria included presence of external or internal resorption of tooth, PDL widening, periapical involvement and teeth showing root apex development.

In order to maintain the same contrast and brightness, all radiographs were captured on standard settings and same angulation using paralleling technique and to reduce the bias of methodology.

Inter examiner agreement

Baseline agreement between the examiners was 0.91 and 0.94 at final follow up which was considered excellent and p value for both baseline and follow up was significant. ($p < 0.05$)

Follow up loss

Two teeth each from Biodentine, Theracal and Dycal groups (10% each) were dropped from the study because the patients failed to report for the follow up (shown in Figure 1).

A total of 54 young permanent molars from 49 patients (23 males and 26 females) with a male to female ratio of 0.88:1.00 were followed up till the end of study. Overall mean age was 12.36 years ±2.29 with a range of 7 to 15 years. Out of 54 teeth, 17(33.33%) teeth were mandibular and 37 (66.66%) were maxillary teeth (Table 1).

Figure 2: Group A (Biodentine)Preoperative (A) and postoperative follow up radiographs (B- 3 weeks, C- 3 months, D- 6 months, E- 12months, F- 18 months, G- 24 months)

Figure 3: Group B (Theracal LC) Preoperative (A) and postoperative follow up radiographs (B- 3 weeks, C- 3 months, D- 6 months, E- 12months, F- 18 months, G- 24 months)

Figure 4: Group C Dycal) Preoperative (A) and postoperative follow up radiographs (B- 3 weeks, C- 3 months, D- 6 months, E- 12months, F- 18 months, G- 24 months)

In the present study, clinical and radiographic signs and symp-

it as a favorable IPT material.

CONCLUSION

Following conclusions can be drawn from present trial:

- The success rate of Calcium hydroxide for IPT is significantly lower in comparison to Theracal, and Biodentine at 24 months follow-up.
- Calcium silicate based materials are better alternatives for IPT in young permanent teeth.
- With improved physical and chemical qualities, calcium silicate materials such as Biodentine and Theracal are proven to be superior to calcium hydroxide.

Disclosure Statement:

The authors deny any conflict of interest related to this study.

Funding sources:

NIL

REFERENCES

1. Stanley HR. Pulp capping: conserving the dental pulp—can it be done? Is it worth it?. Oral Surg Oral Med Oral Pathol;68(5):628-39. 1989.
2. Kidd EM. How 'clean' must a cavity be before restoration?.Caries Res;38(3):305-313. 2004.
3. Ranly DM, Garcia-Godoy F. Current and potential pulp therapies for primary and young permanent teeth. J Dent; 28(3):153-61. . 2000.
4. Makkar S, Kaur H, Aggarwal A, Vashisht R. A confocal laser scanning microscopic study evaluating the sealing ability of mineral trioxide aaggregate, Biodentine and a new pulp capping agent- Theracal. Dental Journal of Advance Studies;3(1):20-25.2015.
5. Garrocho-Rangel A, Quintana-Guevara K, Vázquez-Viera R, Arvizu-Rivera JM, Flores-Reyes H, Escobar-García DM, et al. Bioactive Tricalcium Silicate-based Dentin Substitute as an Indirect Pulp Capping Material for Primary Teeth: A 12-month Follow-up. Pediatr Dent;15;39(5):377-382. . 2017.
6. Taha NA, Abdulkhader SZ. Full Pulpotomy with Biodentine in Symptomatic Young Permanent Teeth with Carious Exposure. J Endod;44(6):932-937.2018.
7. Rajasekharan S, Martens LC, Cauwels RG, Verbeeck RM:Biodentine material characterstics and clinical applications: a review of the literature. Eur Arch Paediatr Dent;15(3):147-58. 2014.
8. Gandolfi MG, Siboni F, Prati C. Chemical–physical properties of Theracal, a novel light-curable MTA-like material for pulp capping. IntEndod J; 45(6):571-579. 2012.
9. Fontana M, Young D, Wolff M, Pitts NB, Longbottom C. Defining dental caries for 2010 and beyond. Dent Clin North Am; 54(3):423-440. 2010.
10. Barthel CR, Rosenkranz B, Leuenberg A, Roulet JF. Pulp Capping of Carious Exposures: Treatment Outcome after 5 and 10 Years: A Retrospective Study. J Endod;26:525-528. 2000.
11. Bjorndal L, Reit C, Bruun G, Markvart M, Kjaeldgaard M.Nasman P et al.Treatment of deep caries lesions in adults: randomized clinical trials comparing stepwise vs. direct complete excavation, and direct pulp capping vs. partial pulpotomy. Eur J Oral Sci; 118:290-297.2010.
12. Dammaschke T, Leidinger J, Schafer E: Long-term evaluation of direct pulp capping—treatment outcomes over an average period of 6.1 years. Clin Oral Investig;14(5):559-567. 2010.
13. Banerjee A, Watson T, Kidd EA. Dentine caries excavation: a review of current clinical techniques. Br Dent J;188:476-482. 2000.
14. Maltz M, Oliveira EF, Fontanella V, Carminatti G. Deep Caries Lesions after Incomplete Dentine Caries Removal: 40- Month Follow-Up Study. Caries Res;41(6):493-496. .2007.
15. Gruythuysen RJ, Van Strijp AJ, Wu MK. Long-term Survival of Indirect Pulp Treatment Performed in Primary and Permanent Teeth with Clinically Diagnosed Deep Carious Lesions. J Endod;36(9):1490-1493. 2010.
16. Bjorndal L, Larsen T, Thylstrup A. A clinical and microbiological study of deep carious lesions during stepwise excavation using long treatment intervals. Caries Res; 31(6): 411-417.1997.
17. Neves A,Coutinho E., De Munck, J, Van Meerbeek, B. Caries-removal effectiveness and minimal-invasiveness potential of caries-excavation techniques: a micro-CT investigation. J Dent;39(2):154-162.2011.
18. Hilton TJ. Keys to clinical success with pulp capping: a review of the literature. Oper Dent;34(5):615-25.2009.
19. Cox CF, Subay RK, Ostro E, Suzuki S, Suzuki SH. Tunnel defects in dentin bridges: their formation following direct pulp capping. Oper Dent; 21(1):4-11.1996.
20. Farhad A, MohammadiZ.Calcium hydroxide: a review. Int Dent J;55(5): 293-301.2005.
21. Leye Benoist F, Gaye NdiayeF, Kane AW, Benoist HM, Farge P. Evaluation of mineral trioxide aggregate (MTA) versus calcium hydroxide cement (Dycal) in the formation of a dentine bridge: a randomised controlled trial. Int Dent J; 62(1):33-9.2012.
22. Sultana R, Hossain M, Alam MS. Evaluation of clinical and radiological outcomes of mineral trioxide aggregate and calcium hydroxide as indirect pulp capping agents in the treatment of deep carious lesion of permanent teeth. Bangabandhu Sheikh Mujib Medical University J; 9:140-145. 2016.
23. Nirschl RF, Avery DR. Evaluation of a new pulp capping agent in indirect pulp therapy. ASDC J Dent Child;50(1):25-30. 1983.
24. Petrou MA, Alhamoui FA, Welk A, Altarabulsi MB, Alkilzy M, H Splieth C. A randomized clinical trial on the use of medical Portland cement, MTA and calcium hydroxide in indirect pulp treatment. Clin Oral Investig;18(5):1383-9. 2014.
25. Mathur VP, Dhillon JK, Logani A, Kalra G: Evaluation of indirect pulp capping using three different materials: A randomized control trial using cone-beam computed tomography. Indian J Dent Res; 27(6):623-629.2016.
26. Orhan AI, Oz FT, Ozcelik B, Orhan K. A clinical and microbiological comparative study of deep carious lesion treatment in deciduous and young permanent molars. Clin Oral Investig;12(4):369-78.2008.
27. Orhan AI, Oz FT, Orhan K. Pulp exposure occurrence and outcomes after 1-or 2-visit indirect pulp therapy vs complete caries removal in primary and permanent molars. Pediatr Dent; 32(4):347-55. 2010.
28. Maltz M, Alves LS, Jardim JJ, Moura Mdos S, de Oliveira EF.Incomplete caries removal in deep lesions: A 10-year prospective study. Am J Dent; 24(4):211-214.2011.
29. Zanini M, Sautier JM, Berdal A, Simon S: Biodentine induces immortalized murine pulp cell differentiation into odontoblast-like cells and stimulates biomineralization. J Endod;38(9):1220-6. 2012.
30. Hashem D, Mannocci F, Patel S, Manoharan A, Brown JE, Watson TF, ET AL.Clinical and radiographic assessment of the efficacy of calcium silicate indirect pulp capping: a randomized controlled clinical trial. J Dent Res;94(4):562-8. 2015.
31. Arandi NZ, Rabi T. TheraCal LC: From biochemical and bioactive properties to clinical applications. Int J Dent;34.846-53. 2018.
32. Camilleri J, Laurent P, About I. Hydration of biodentine, theracal LC, and a prototype tricalcium silicate-based dentin replacement material after pulp capping in entire tooth cultures. J Endod;40(11):1846–1854.2014.
33. Poggio C, Lombardini M, Colombo M, Beltrami R, Rindi S. Solubility and pH of direct pulp capping materials: a comparative study. Journal of Applied Biomaterials & Functional Materials;13(2):181-185.2015.
34. Yamamoto S, Han L, Noiri Y, Okiji T. Evaluation of the Ca ion release, pH and surface apatite formation of a prototype tricalcium silicate cement. Int Endod J;50(2):73-82. 2017.
35. Gandolfi MG, Siboni F, Botero T, Bossù M, Riccitiello F, Prati C. Calcium silicate and calcium hydroxide materials for pulp capping: biointeractivity, porosity, solubility and bioactivity of current formulations. J ApplBiomaterFunct Mater;13(1):43-60. 2015.